Cloud-Based Collaborative Document Editing and Version Control Platform

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Cloud Computing Capstone Project

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Introduction

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- Cloud-based collaboration enables multiple users to access and edit documents from different locations.
- Traditional document sharing can create **duplicate files, conflicting edits, and difficulty tracking changes**.
- Real-time collaborative editing allows users to work on the **same document simultaneously**.
- The proposed system combines **real-time editing with cloud-based version control**.
- **Operational Transformation (OT) / CRDT** techniques are used to handle concurrent edits and maintain consistency.
- **WebSocket communication** provides fast, continuous synchronization between users.
- A version-control module maintains **complete document history**, enabling users to review and retrieve previous versions.

Objectives

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- Enable real-time collaboration between multiple users on the same document.
- Minimize synchronization latency during collaborative editing.
- Resolve concurrent editing conflicts accurately using OT/CRDT techniques.
- Support multiple concurrent editors without significant performance degradation.
- Maintain complete version history of document changes.
- Provide fast version retrieval and restoration of previous document states.
- Ensure secure and reliable cloud-based document storage and access.

Key Performance Parameters

- Sync Latency
- Conflict Resolution Accuracy
- Concurrent Editor Capacity
- Version Retrieval Time

Literature Review / Background

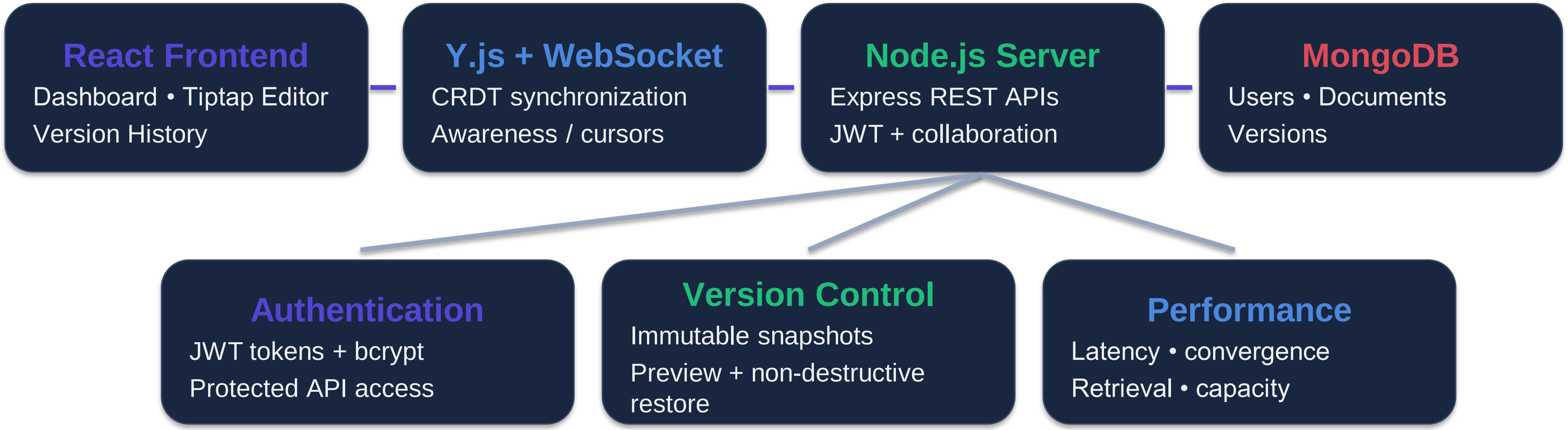
Study / Technology	Main Contribution	Relevance to Our Project
Ellis & Gibbs (1989)	Introduced concepts for concurrency control in groupware systems	Provides the foundation for collaborative editing
Lamport (1978)	Introduced logical clocks and event ordering in distributed systems	Helps understand ordering of concurrent operations
Operational Transformation (OT)	Transforms concurrent operations to maintain document consistency	Used for conflict resolution
CRDTs	Enables distributed data to merge while maintaining eventual convergence	Alternative approach for real-time collaboration
WebSocket (RFC 6455)	Provides persistent, two-way client-server communication	Enables real-time document synchronization
Version Control Systems	Maintain revisions and allow historical recovery	Supports document version history and restoration

Methodology

- Technology Stack
- Frontend: React.js, Vite, Tiptap / ProseMirror
- Real-time Sync: Y.js CRDT + WebSocket
- Backend: Node.js + Express
- Database: MongoDB / MongoDB Atlas-ready
- Security: JWT + bcrypt password hashing
- Version Control: MongoDB immutable version snapshots

Development Flow:

1. Authenticate user → 2. Create/open document → 3. Connect Y.js provider → 4. Exchange edits through WebSocket → 5. Persist latest content → 6. Save/restore versions → 7. Run performance benchmarks.



Edit → Y.js CRDT update → WebSocket broadcast → connected clients converge → latest content persisted → version snapshot saved on demand

Engineer to Excel

Implementation

Module 1

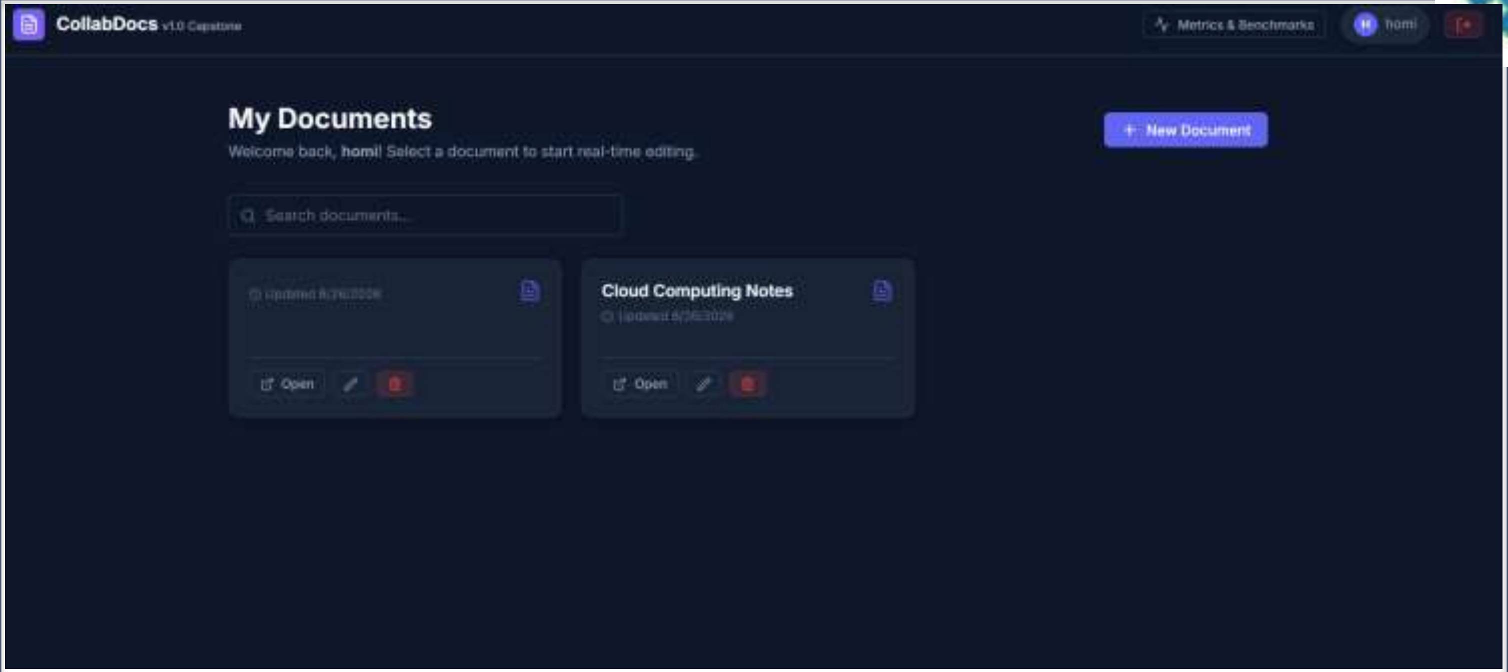
Real-time collaborative rich-text editing with Tiptap, Y.js, WebSockets and collaborator awareness.

Module 2

Version snapshots, history timeline, preview and non-destructive restore in MongoDB.

Security

JWT-protected APIs, bcrypt password hashing and document access control.



Dashboard: document creation, search, open, rename and delete controls

Results & Discussion

Engineer to Excel

Observed benchmark run

- Low WebSocket round-trip time supports responsive collaboration.
- 100% convergence indicates the controlled Y.js concurrent-edit test reached a consistent state.
- Version retrieval was measured through the real API/database path.
- The current validated capacity is 5–20 concurrent users; larger-scale deployment should be load-tested separately.
- Note: values are measurements from the demonstrated benchmark run, not fixed guarantees.

Challenges & Limitations

- Correct Y.js provider/awareness lifecycle: improper initialization can crash the editor.
- Reliable concurrent synchronization requires careful WebSocket and CRDT lifecycle management.
- Persistent content and immutable versions must remain separate from transient CRDT state.
- Document privacy requires backend authorization, not only frontend filtering.
- Current performance results are local/development measurements; production cloud scaling needs further load testing.
- MongoDB Atlas/cloud deployment and multi-instance WebSocket scaling require additional production configuration.

Future Scope

- Deploy frontend, backend and MongoDB Atlas in a production cloud environment.
- Add invitation-based sharing, roles and fine-grained document permissions.
- Introduce Redis/pub-sub or a managed collaboration service for multi-server WebSocket scaling.
- Add offline-first editing and automatic reconnection with durable CRDT persistence.
- Add document comments, mentions, notifications and richer collaboration analytics.
- Add encryption-at-rest/key management, audit logs and stronger enterprise security controls.

Conclusion

- The project demonstrates a cloud-ready collaborative document platform that combines:
- Real-time editing using Tiptap + Y.js CRDT
- WebSocket-based synchronization and collaborator awareness
- MongoDB persistence and immutable version history
- JWT authentication and protected backend services
- Non-destructive version restoration
- Empirical performance benchmarking
- Overall, the system provides a practical foundation for collaborative cloud applications while demonstrating the key concepts of synchronization, conflict resolution, persistence and scalability.

References

- Fasola, O.S. and Abimbola, M.O., 2023. Collaborative technology for information sharing, knowledge creation and management in libraries. *Gateway Information Journal*, *24*(1 & 2), pp.33-46.
- Castelló, M., Kruse, O., Rapp, C. and Sharples, M., 2023. Synchronous and asynchronous collaborative writing. In *Digital writing technologies in higher education: Theory, research, and practice* (pp. 121-139). Cham: Springer International Publishing.
- Rodriguez, B.A., Aquiatan, V.T., Verallo, C.J.A., Agpad, S.B., De Loyola, R.C. and Bibangco, E.J.P., 2024, June. Digitalization of document management and monitoring in the Department of the Interior and Local Government Negros Occidental. In *2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)* (pp. 1-8). IEEE.
- Hammad, M., Iqbal, J., Hassan, C.A.U., Hussain, S., Ullah, S.S., Uddin, M., Malik, U.A., Abdelhaq, M. and Alsaqour, R., 2023. Blockchain-based decentralized architecture for software version control. *Applied Sciences*, *13*(5), p.3066.
- Chen, K.Y., Toro-Moreno, M. and Subramaniam, A.R., 2025. GitHub is an effective platform for collaborative and reproducible laboratory research. *ArXiv*, pp.arXiv-2408.
- Scroggie, K.R., Burrell-Sander, K.J., Rutledge, P.J. and Motion, A., 2023. GitHub as an open electronic laboratory notebook for real-time sharing of knowledge and collaboration. *Digital Discovery*, *2*(4), pp.1188-1196.

Thank You!